



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

SYMPHONY PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 Q&A on DI, routing, Doctrine, events, security, and testing

TOTAL: 10 QUESTIONS

Q1 What is Symfony Flex and how does it manage bundles?

Threat Model

Q6 How does Symfony Security's firewall system work?

Observability

Q2 How does Symfony's DI container and autowiring work?

Architecture

Q7 How do Symfony Forms handle validation?

Lifecycle

Q3 How does Symfony's routing system work?

Configuration

Q8 How do you create a Symfony Console command?

Compliance

Q4 What is Doctrine ORM and how does it differ from Active Record?

Hardening

Q9 How does Symfony Cache work with PSR-6?

Testing

Q5 How does the Symfony EventDispatcher work?

SSO / IdP

Q10 How do you write functional tests in Symfony?

Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Symfony Flex is a Composer plugin

Mitigation

Symfony Flex is a Composer plugin that maps package aliases (symfony/orm-pack) to recipes stored on a Flex server. On install it runs the recipe: creates config files, registers the bundle in bundles.php, and adds env vars to .env. This automates all boilerplate.

A6 A firewall in security.yaml defines which

Observability

A firewall in security.yaml defines which URLs it protects and how (form_login, jwt, api_key). The security layer authenticates the request and populates a token in the TokenStorage. Access control rules (access_control) check roles on top of the authenticated token.

A2 Symfony reads services.yaml and scans src/

Primitives

Symfony reads services.yaml and scans src/ for classes to register as services. Autowiring reads constructor type hints and injects matching services automatically. Interface bindings (\$logger: Psr\Log\LoggerInterface) resolve abstractions to concrete implementations.

A7 A FormType defines fields and maps

Automation

A FormType defines fields and maps them to a model class. \$form->handleRequest(\$request) binds submitted data. \$form->isSubmitted() && \$form->isValid() checks Data Annotation (or YAML) constraints. \$form->getErrors(true) returns validation messages.

A3 #[Route('/users/{id}', name: 'user_show',

Hardening

#[Route('/users/{id}', name: 'user_show', methods: ['GET'])] on a controller method registers the route via PHP 8 attributes. Alternately use config/routes.yaml. Route::generate('user_show', ['id'=>1]) produces the URL. Prefix routes with #[Route('/api')] on the class.

A8 Extend Command. Set the name in

Governance

Extend Command. Set the name in configure() via \$this->setName('app:greet'). Define input arguments/options with addArgument/addOption. Implement execute(\$input, \$output). Run with php bin/console app:greet --name=Alice. Tag with console.command for auto-discovery.

A4 Doctrine uses the Data Mapper pattern:

Gotchas

Doctrine uses the Data Mapper pattern: entities are plain PHP objects with no database knowledge. An EntityManager (separate object) handles persistence. Active Record (e.g., Eloquent) embeds DB logic in the model itself. Data Mapper enables richer domain models.

A9 Symfony's CacheInterface (a superset of PSR-6)

Assessment

Symfony's CacheInterface (a superset of PSR-6) provides get(\$key, callable). Use cache.adapter.redis in config to back the pool with Redis. @Cache(expires='1 hour') on a controller action caches the HTTP response via the HttpCache reverse proxy.

A5 Dispatch an event: \$dispatcher->dispatch(new

Federation

Dispatch an event: \$dispatcher->dispatch(new OrderCreated(\$order)). A subscriber implements EventSubscriberInterface and returns getSubscribedEvents(): [OrderCreated::class => 'onOrderCreated']. Tag the subscriber with kernel.event_subscriber for auto-registration.

A10 Extend WebTestCase. \$client =

Optimization

Extend WebTestCase. \$client = static::createClient(); \$crawler = \$client->request('GET', '/users/1'); \$this->assertResponseIsSuccessful(). Use \$client->clickLink() and \$client->submitForm() to simulate user interaction. KernelTestCase is used for service-layer tests.